Case 1: Manufacturing quality control

Problem Statement:

The manufacturing company uses an AI-based computer vision system to inspect electronic components, classifying each product as **defective** or **non-defective**. The quality-control engineer needs to answer four key questions about the defect process:

1. What is the probability of finding a particular number of defective products in batch?
2. How many defects should the company expect?
3. How does the defect probability affect the distribution?
4. How many inspections might be needed before the first defect is detected?

Solution:

Given data:

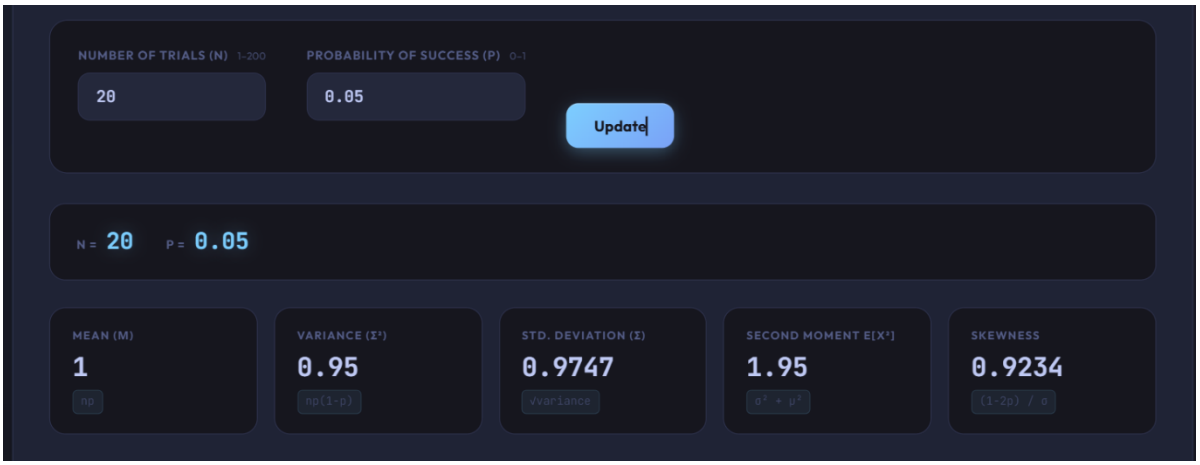
1. Assumed defect probability: p = 0.05.
2. Batch size: n = 20.
3. Total inspected: 200.
4. Total observed defects: 11
5. The assumed model parameter p = 0.05.

Sample batch data:

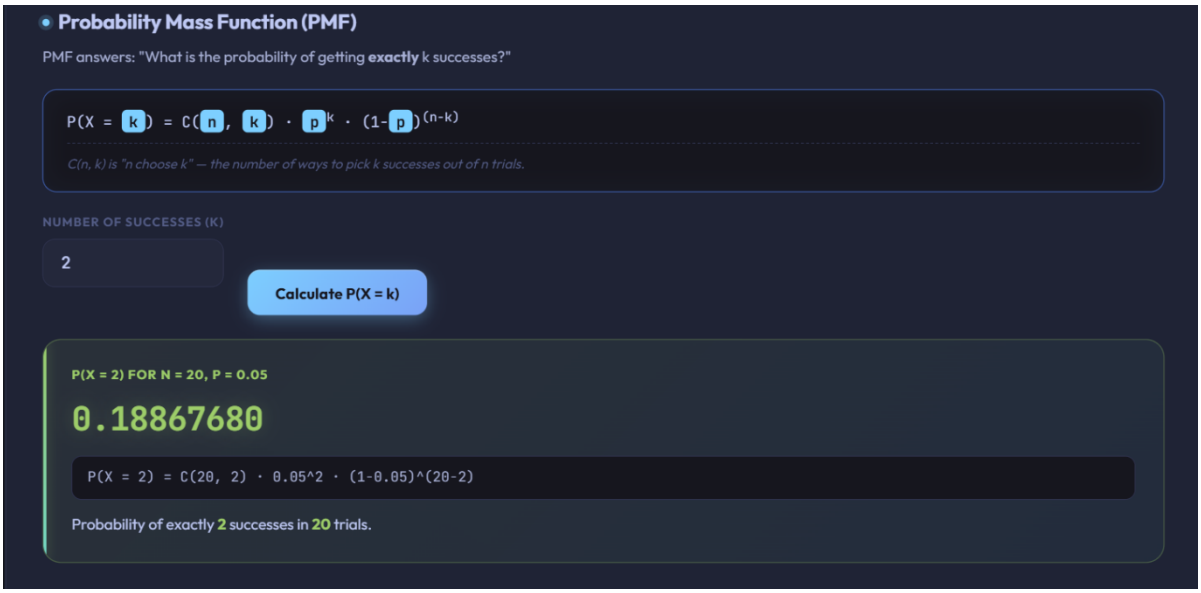
Batch	Products inspected	Defective products observed
1	20	1
2	20	0
3	20	2
4	20	1
5	20	3
6	20	0
7	20	1
8	20	2
9	20	1
10	20	0

Now lets solve the given questions using different probability distributions.

1Q What is the probability of finding a particular number of defective products in a batch?



In this we have to find the number of defective products. So, we use the formula of probability mass function.



The probability of finding exactly 2 defective products in a single batch of 20 is 0.1887.

2Q How many defects should the company expect?

We need to calculate mean and variance.



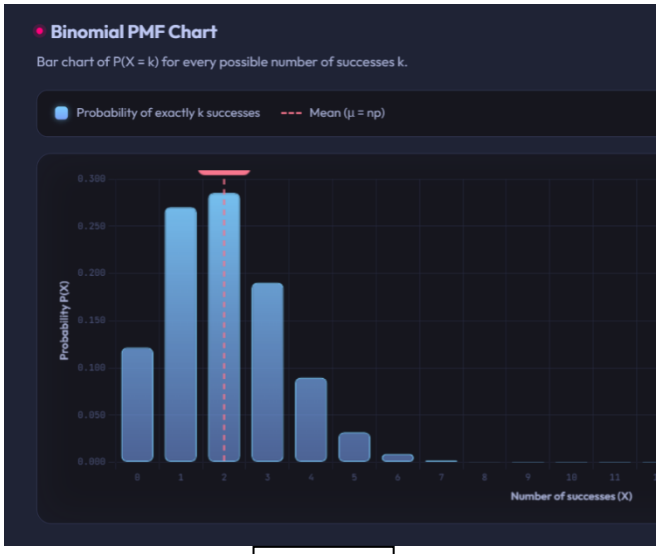
1. The company should expect to find 1.0 defective product per batch of 20 items.
2. The variance of the defect distribution is 0.95.

3Q How does the defect probability affect the distribution?

To check the defect probability, we have to compare the original defect probability (p= 0.05) against an increased defect probability (p = 0.10).



p=0.05

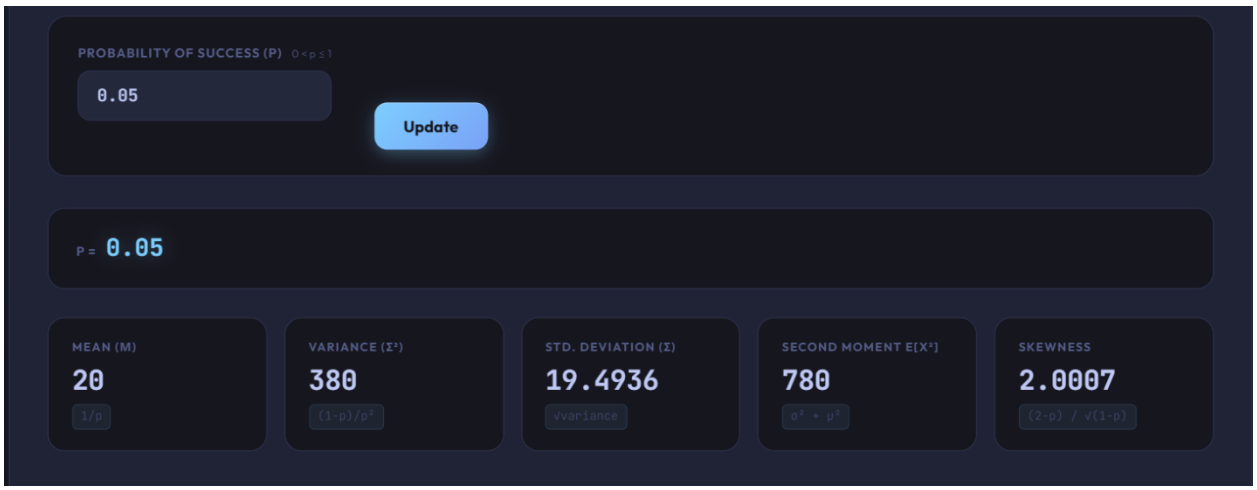


p=0.10

1. When we change the p value the center of the distribution shifts toward the right.
2. The distribution becomes wider and spreads out more.

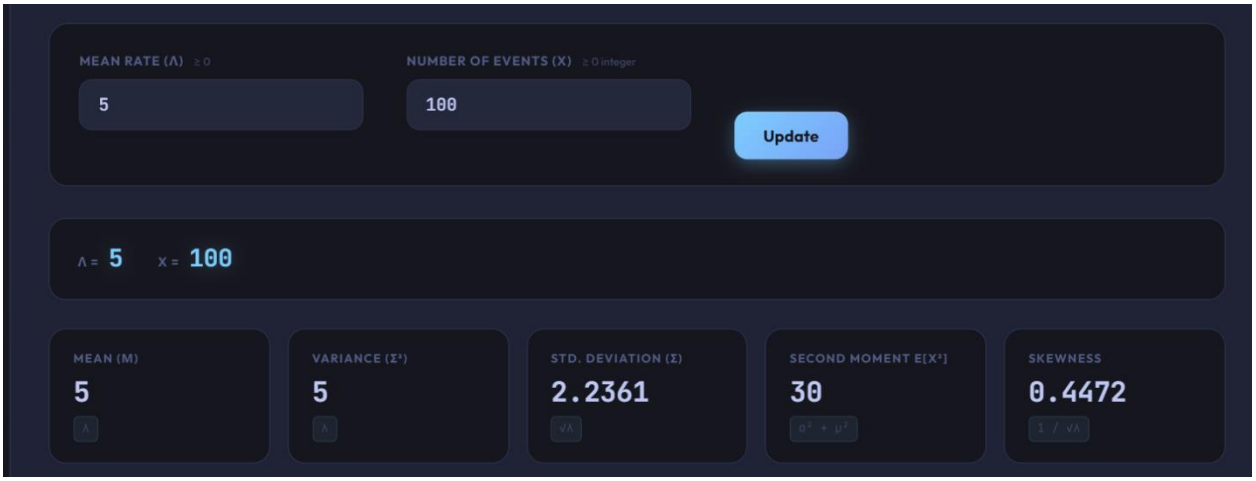
4Q How many inspections might be needed before the first defect is detected?

We use geometric distribution to solve this question and the mean gives the number of inspections which might be needed before the first defect.



Poisson’s Distribution:

- Number of Trials (n): 100
- Poisson Rate Parameter (lambda): 5



USE CASE 2: CLOUD OPERATIONS AND CYBERSECURITY MONITORING

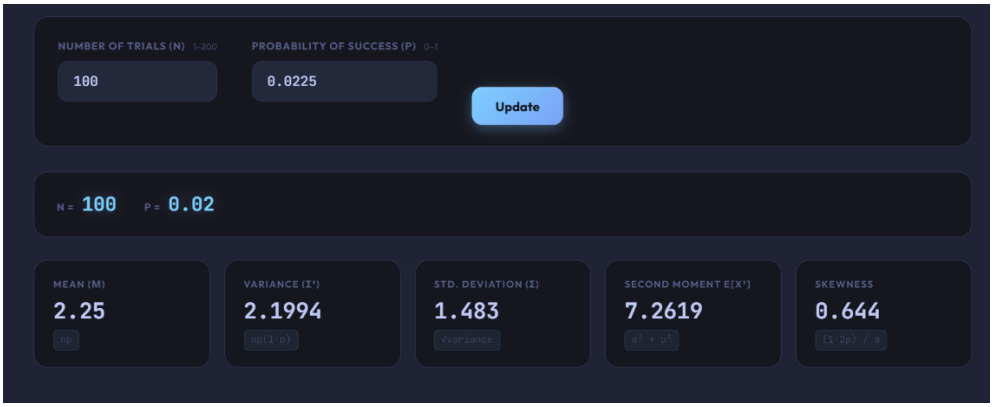
Problem Statement:

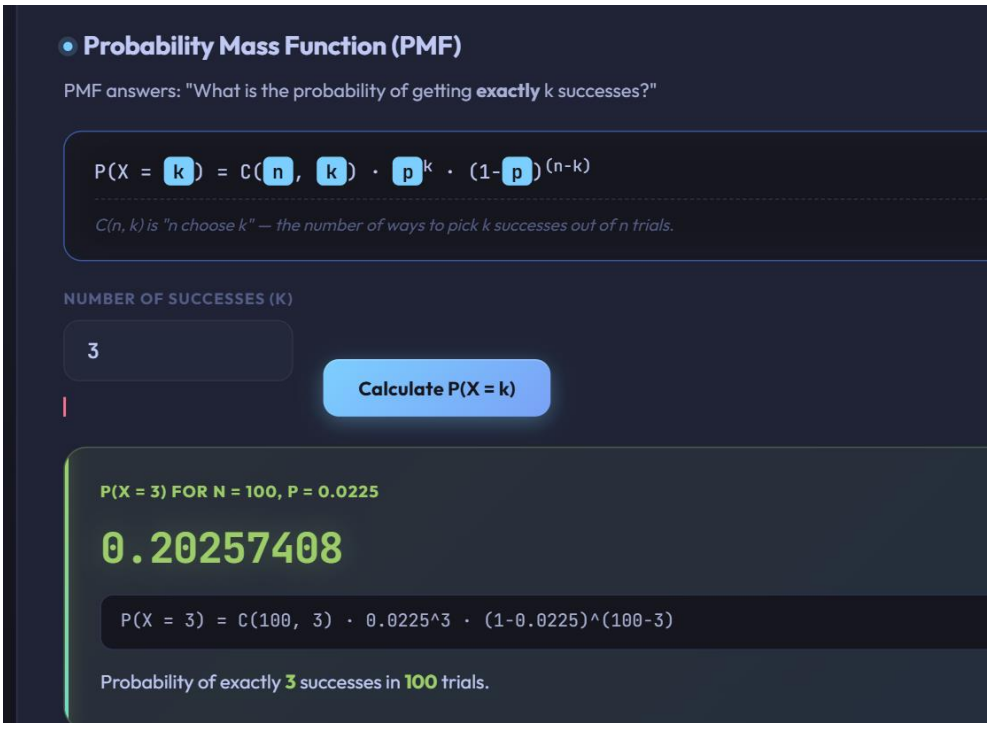
A cloud service provider operates servers and APIs used by thousands of customers. Its operations team monitors failed requests, security alerts, and retry attempts.

Step 1: Enter the data.

Step 2: Select Binomial

- Action: Set *n*=100 and p=0.0225.
- Task: Calculate the probability of exactly 3 failures (k=3) and view the PMF (Probability Mass Function).





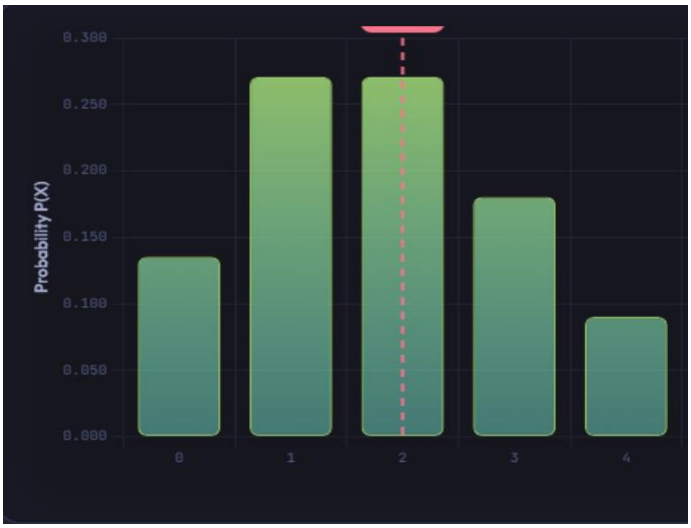
Conclusion: There is a 20.31% chance that exactly 3 out of 100 requests will fail.

Step 3: Select Poisson

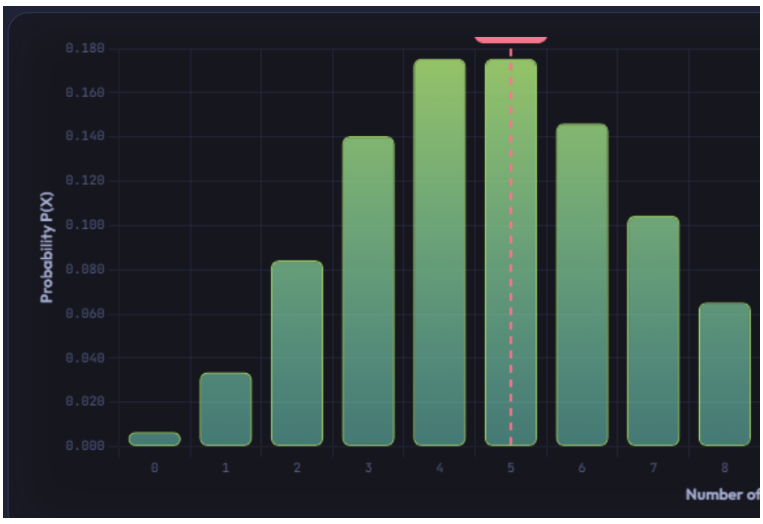
- Action: Set λ=3.
- Task: Calculate the probabilities of 0, 1, 2, 3, and higher alert counts.



Step 4: Experiment with λ



λ = 2

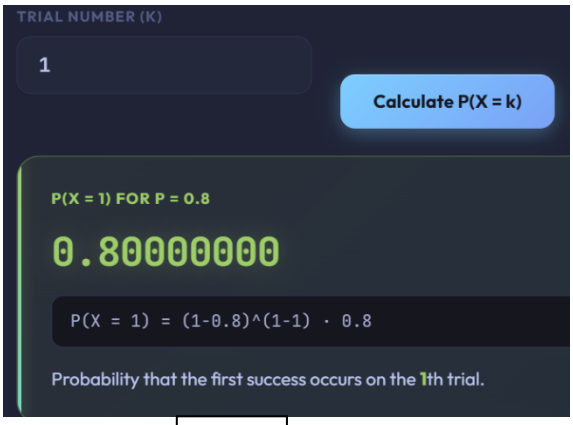


λ = 5

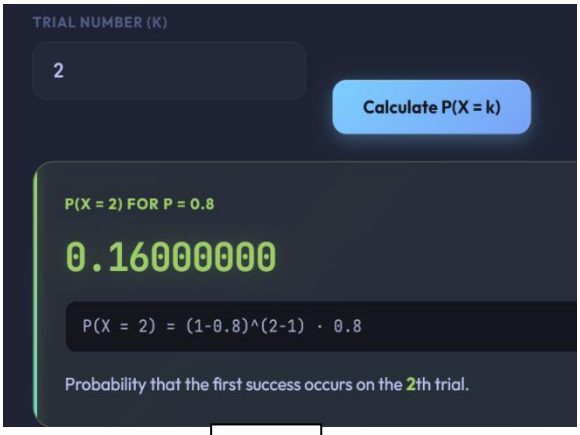
Conclusion: As λ increases, the center of the graph shifts to the right and the distribution becomes wider and more symmetrical

Step 5: Select Geometric

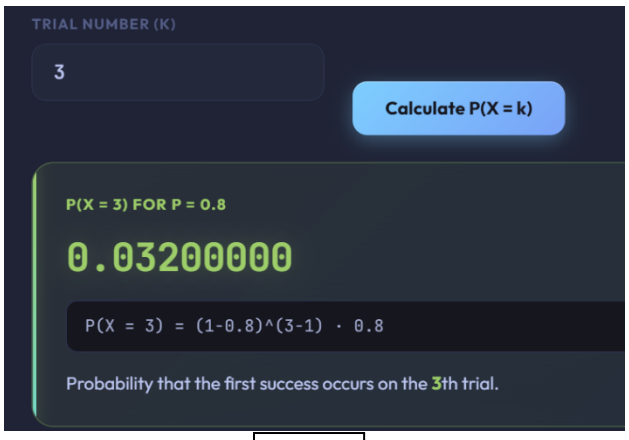




K=1



K=2



K=3

Step 6:

1.How many requests are likely to fail in a fixed set of requests?

We use the Binomial distribution because we have a fixed number of independent requests There is a 20.31% probability that exactly 3 requests out of 100 will fail.

2.How many security alerts might occur during a fixed time interval?

We use the Poisson distribution because we are looking at the number of random events happening in a fixed amount of time .The expected average number of security alerts in a 10-minute window is 3.

3.How many retry attempts might be required before a request succeeds?

We use the Geometric distribution when we want to know the number of trials needed to get the first success . A request succeeds in 1.25 attempts on average, meaning it almost always works immediately.